

MASTER THESIS



Model based Design of a Sailboat Autopilot

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Theophil Ruzicka: *Model based Design of a Sailboat Autopilot*, © June 12, 2017

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ABSTRACT

Autopilots for sailboats are useful assisting tools for sailors. Commercially available autopilots are not able to actuate the sails. This means that actual autopilots can not control the sails actuation while the boat keeps the desired course. The idea of this thesis project is to enable sail actuation for autopilots in order to control the sail angle. Sailboat autopilots with the ability to actuate the sails are able to control the sail angle, depending on changing wind directions and velocity.

In this thesis project a sailboat autopilot with integrated sail control has been implemented. A physical prototype, a sailing model boat, shows the feasibility of the sail angle control. The electronic development of this thesis project focuses on compactness and fail safety. Special attention has been given to the development of the control system which is responsible for keeping the desired course and the sail control. The control system is a combination of learning and feedback control. Learning is used to compute the sail angle and feedback control is used to follow the desired course.

The final experiment shows the feasibility of sailboat autopilots with sail control. But exposed that speed measuring errors can have an impact on the accuracy of the system. The tests were done with the physical prototype on a lake near Halmstad.

It can be concluded that sail control for sailboat autopilots may be an extension for existing systems on sailboats. Another benefit of this thesis is, that it can be used as a platform for studying different problems in control. A possible future step is, to push this thesis project to a product. Therefore it would be needed to extend the experiments in order to get more evaluation data and improve the accuracy by using sensor fusion.

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ACRONYMS

RC Remote Control

PWM Pulse Width Modulated

IC Inter-Integrated Circuit

GPS Geographical Positioning System

NMEA National Marine Electronics Association

RMC Recommended Minimum Sentence C

PID Proportional Integral Derivative

PD Proportional Derivative

PI Proportional Integral

EM Electro Magnetic

INTRODUCTION

In former times sailing ships had a huge relevance for mankind. Sailboats were mostly used for fishing, trading, transportation or even for fighting wars. The use of wind, in order to move ships dates back to the prehistoric age. First attempts of building sailboats were discovered on paintings in Egypt, dating back 5000 years BC[48]. Humans sailed with their boats the seas, before they understood the physics a sailboat uses. At the very beginning, sailing was all about transportation, it was a tool people used to ferry commodities from one point to another. It was a tool for people.

Nowadays, the priority for sailors moved from transportation to sport or vacation. For some people, sailboats are more than just a boat, they are their passion, which they use to participate in sailing races. There are sailing competitions, like the Volvo Ocean Race¹, or the famous America's Cup². But sometimes, sail boats are used for an enjoyable vacation only and they are maneuvered by hobby yachtsmen. Most times, those sailors are no experts, so they would use the on-board autopilot.

A sailboat can be actuated by two devices, the rudder and the sails. The rudder floats through the water and steers the boat by using hydrodynamic forces. The sails use the power of the wind to move the boat forward and to speed it up, they are built to use the winds aerodynamic optimally. Sails are producing thrust by using aerodynamic forces. The procedure of adjusting the angle of the sails is called sail trim, if the forces in the sails do not stream in the optimal way, it can result in unusual sailing behavior. One way of reducing such behavior is to use an autopilot, they are a very useful assisting tool, especially for novice sailors.

Commercial autopilots control the rudder only, they are not able to control the sail angle. This is already a useful functionality for sailors, but not a proper autopilot. It must also be able to adjust the sails automatically and to react on changing conditions, like a change in the wind stream, a wrong sail position or an unusual sailing behavior. Extending those existing autopilot systems, by adding knowledge about sailing and enabling sail control, is going to improve them.

¹ more detail see: www.volvooceanrace.com/en/home.html

² more detail see: www.americascup.com/

1.1 RELATED WORK

1.1.1 Literature Review

The main reference for the literature review is the domain of sailing robots.

In [28] it is shown how a simple controller for a sailboat is implemented. This approach uses a proportional controller to actuate the rudder. In order to avoid a swinging behavior of the system, the controller can be extended with an integrative and a derivative term as it is illustrated in [22].

For controlling the sails, there are some different approaches. In [37] it can be seen how the sail angle is computed by using complex mathematical models of the sails and the boat. This approach is difficult to use on different sailboats because therefor it is necessary to know the exact models which describes the behavior of the sails and the boat. In order to gain a more flexible control for a bigger range of sailboats, fuzzy logic can be used, this approach is described in [13, 44]. Another possible approach to control the sail angle would be to use machine learning algorithms. The use of machine learning for the sail angle computation could reduce the processing power in the system and make the system independent to any mathematical models of the sails and the boat.

[26, 31] shows how it is possible to measure the wind direction with a custom-made sensor. This is a good approach to build a wind sensor for this thesis project.

1.1.2 Related Products

Commercially available autopilots for sailboats can steer the rudder only. They are used to stay on an accurate course from 0° to 360° , or to follow the current wind stream by measuring the wind direction with a wind sensor (called vane). Table 1 explains briefly the key features of various commercially available autopilot systems for sailboats, with the focus on their functionality and their required input. The header of the Table 1 displays the major producer for sailboat autopilots and their most common products. The first column displays their features and which data they access. As the table below describes, there are products available which can control the rudder by using the current heading and the wind direction. But so far, there is no system which is able to control the rudder and the sails at the same time. Furthermore, they do not have special knowledge about the sailing theory or the physics needed for sailing, which are the key factors to enable sailing control depending on lee or weather helm.

Table 1: Comparison of existing autopilot systems

	Raymarine ³	SI-TEX ⁴	Furuno ⁵	Humminbird ⁶	B&G ⁷	Simrad ⁸	This thesis
Rudder Actuation	✓	✓	✓	✓	✓	✓	✓
Sail Actuation	x	x	x	x	x	x	✓
Heading (GPS/Compass)	✓	✓	✓	✓	✓	✓	✓
Wind Vane	✓	x	x	✓	✓	✓	✓
Lee / Weather helm	x	x	x	x	x	x	✓

1.1.3 What is lee helm and weather helm?

A sailboat can either be lee helm, weather helm or neutral. Lee or weather helm is a sailing behavior occurring when a sailboat is on an upwind course. Each behavior appears when the combined forces from the sails and the boat are causing a rotation.

Weather helm [5] means that the forces in the sails are creating a tendency for the boat to turn towards the source of wind. In this situation it can happen that the boat crosses the wind, this can be caused by a wrong sail angle, a changing wind direction or wide roll angle from the boat.

Lee helm [5] means that the sailboat has a tendency to turn away from the source of wind. During this situation it can happen that boat turns too far away and crosses the wind with its back, this can be caused by a wrong sail angle or by a changing wind direction. The principal behavior of lee and weather helm is shown in figure [Figure 1](#) and described with more detail in section [Section 2.3](#).

³ more detail see: www.raymarine.com/view/?id1264

⁴ more detail see: www.si-tex.com/item/sp110autopilotssystem

⁵ more detail see: www.furunousa.com/products/productdetail.aspx?product=NAVPILOT+711C

⁶ more detail see: www.humminbird.com/Products/SC-110-AP-Kit/

⁷ more detail see: www.bandg.com/en-GB/Product-Groups/Triton2/

⁸ more detail see: www.simrad-yachting.com/en-US/Products/Instruments/IS42

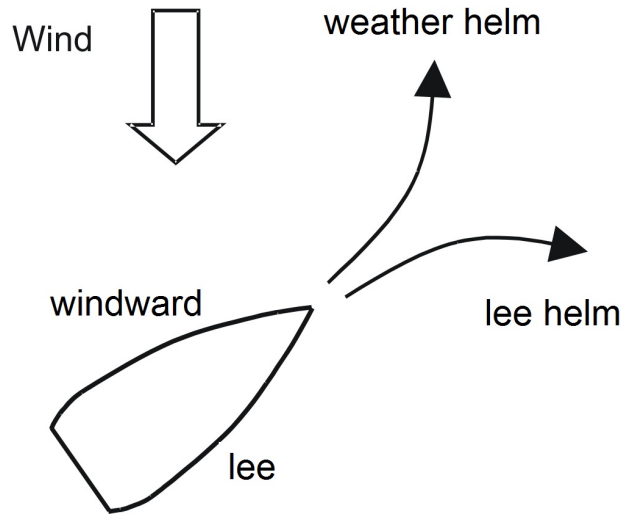


Figure 1: Principals of lee and weather helm

1.2 PROBLEM DEFINITION

Existing autopilot systems for sailboats use either the current heading or the wind direction to control the rudder of a boat. In order to control the sailing behavior by adjusting the sails automatically, it is necessary to calculate the sail angle. The angle of the sail is important for the sailing behavior on a sailboat. The goal of this thesis is to combine the automated sail control and the rudder control. So far, there is no autopilot fitting to this requirement. This leads to the thesis that, **automated sail actuated control is feasible for sailboat autopilots.**

If the sail angle is not optimal, or if the wind direction changes, then the boat can be either lee or weather helm, assuming that the boat is on an upwind course. The upwind course denotes the course where the boat travels diagonally to the wind direction, this maneuver is also called tack. Tack is a sailing maneuver where the lee and windward side of the boat are changing. [Figure 2](#) illustrates an upwind course with tack maneuvers. The remaining question is, **can the sails of a sailboat be adjusted automatically while keeping the desired course?**

Within this thesis it will be shown that it is possible to actuate the sails angle in order to increase the boat speed. This sail control is adjusting the sails while the boat follows the desired course. The feasibility of this thesis will be provided by using a physical model. This model is a small scaled model sailboat, which fits to all the criteria described in this thesis.

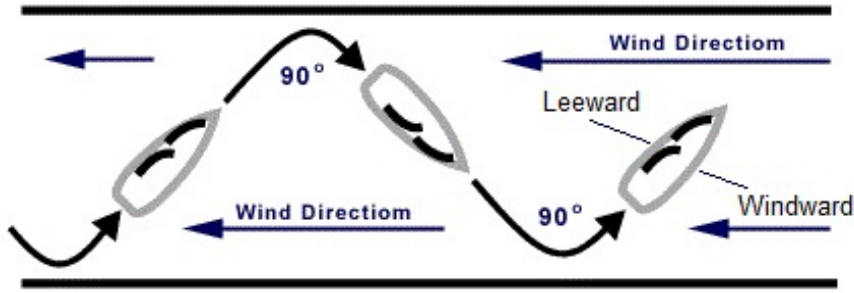


Figure 2: Tack maneuvers on an upwind course [11]

1.3 THESIS CONTRIBUTION

This thesis shows how the manual sail trim can be operated fully automated by the autopilot system. The physical prototype for this thesis is based on related work on sailing robots, which can be found in [29, 43, 42]. The physical prototype is used to demonstrate the feasibility of this thesis. By enabling the sails actuation for autopilots, this thesis increases the usability for sailboat autopilots. This thesis can also be used as a platform for studying different problems in control.

1.4 OUTLINE OF THE THESIS

The following [Chapter 2](#) briefly introduces the basic physics of sailing, the behavior of lee helm and weather helm of a sailboat during an upwind course and the determination for the sailboats behavior.

[Chapter 3](#) presents the complete hardware design process, starting with basic assumptions and requirements and a power consumption diagram. Further all needed sensors for the control systems are described in this chapter.

[Chapter 4](#) describes the implemented control system including the course control and the sail control. The section about the sail control system also describes the optimization for the sail angle.

[Chapter 5](#) illustrates the experiments and tests with the physical prototype and evaluates the achieved results.

Finally [Chapter 6](#) concludes with the summary of this thesis results. A brief review of the outcomes of the developed process is done and potential future extensions and upgrades are suggested.

PHYSICS OF SAILING

The physics of sailing and the behavior of a boat are a combination of hydrodynamic and aerodynamic forces. This chapter describes, how a sailboat uses physical forces in order to produce thrust. It is also explained how various forces affect the behavior of the sailboat. Depending on the wind direction, a sailboat can sail several courses. All possible courses are described in this chapter.

2.1 FORCES AT SAILING

Sailing starts with the force of the wind in the sail. The force produced by the wind stream is divided into the aerodynamic lift force $\vec{L}$ and the aerodynamic drag force $\vec{D}$. The force $\vec{L}$ is perpendicular to the wind direction. $\vec{D}$ is the force which points in the direction of the wind. The vectorial sum of $\vec{L}$ and $\vec{D}$ is the total aerodynamic force $\vec{F}_T$. Equation 1 shows the total aerodynamic force[47].

$$\vec{F}_T = \vec{L} + \vec{D} \quad (1)$$

$\vec{F}_T$ can be segmented into the driving force $\vec{F}_R$ and the heeling force $\vec{F}_H$. $\vec{V}_A$ is the relative velocity of the wind, where α is the apparent wind angle.

Figure 3 describes the forces in the sail.

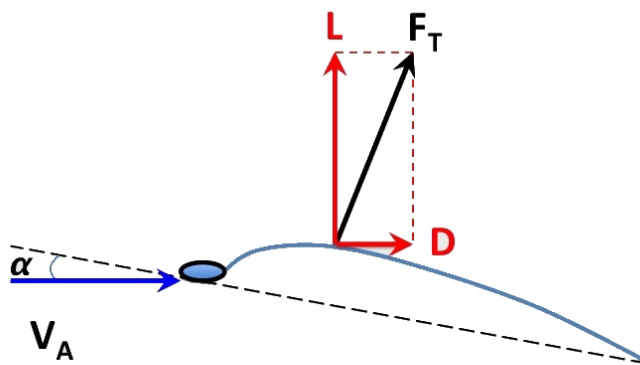


Figure 3: Total forces on a sail[4]

The corresponding force to $\vec{F}_T$ is the lateral force $\vec{F}_{LAT}$. $\vec{F}_{LAT}$ is composed of the hydrodynamic side force $\vec{F}_T$ and the force on the keel, $\vec{F}_R$ which is parallel to the motion of water. $\vec{V}_B$ is the boat velocity. $\vec{V}_T$ is the true velocity of the wind. Figure 4 displays the combined forces from the sail and the lateral force and Equation 2 shows the calculation for the lateral force[35].

$$\vec{F}_{LAT} = \vec{F}_T - \vec{F}_R \quad (2)$$

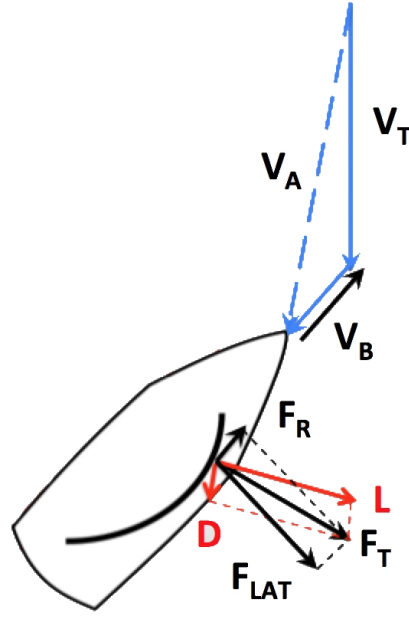


Figure 4: Hydrodynamic forces [1]

2.2 SAILING COURSES

In sailing theory there are two different types of sailing courses, the upwind course and the downwind course. An upwind course is a course where the wind comes from the bow of the boat, whereas the wind blows from the rear at a downwind course. Figure 5 displays all various sailing courses of a sailboat, better known as the points of sail [10].

As it can be seen in Figure 5, the area 'in irons' is impossible to sail, because the boat would travel directly against the wind source. The angle between the two close hauled courses is the area where the sailboat can not sail.

When the wind direction is blowing perpendicular to the sailboats course, it is called beam reach course.

The beam reach is exactly between the downwind course and the upwind course. This is also illustrated in [Figure 5](#).

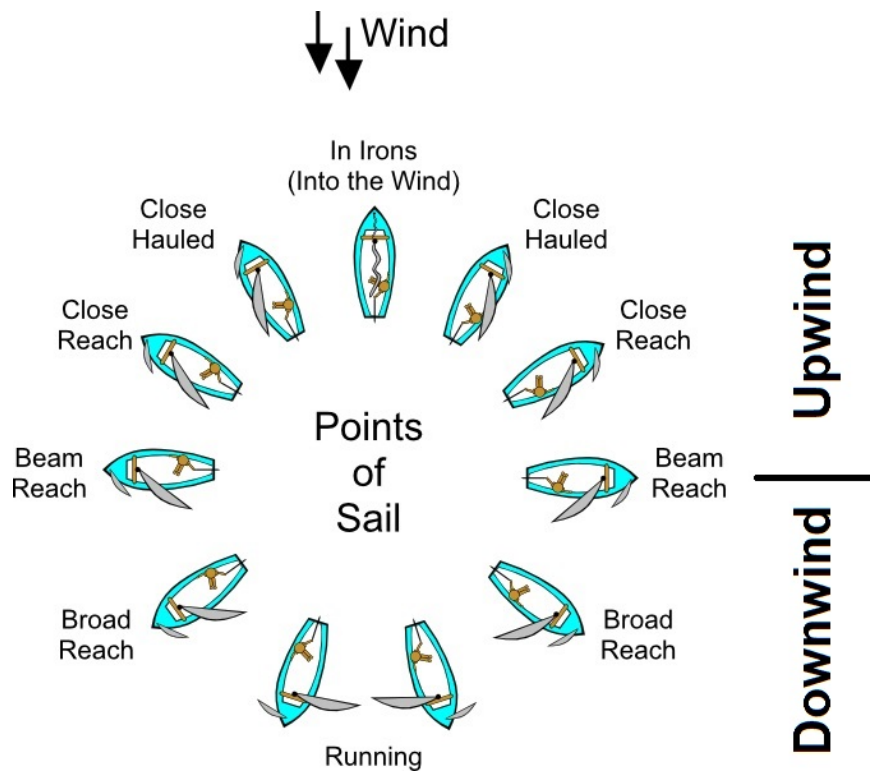


Figure 5: Point of sail[10]

2.2.1 Upwind course

The two downwind courses are 'close hauled' and 'close reach'. The close hauled and close reach. For both courses the wind source comes from the bow. To sail an upwind course, some maneuvers need to be executed, in order to reach the the target position. The maneuvers on an upwind course are called, tack. Tack is a maneuver where the boat crosses the wind source.

This means, if the starboard side of the boat is windward before the maneuver, it switches to leeward. More detail can be seen in [Figure 2](#).

2.2.2 Downwind course

Is the target position of the desired course in the same direction than the wind direction, the course is called downwind. It means, the wind comes from the rear of the boat. The two downwind courses are 'broad reach' and 'running', as shown in [Figure 5](#).

Also while sailing on a downwind course, some maneuvers need to be executed. The maneuvers on the downwind course are called jib.

This means, the starboard side of the boat is the windward side before the maneuver, it is the starboard side afterwards. The jib maneuver is illustrated in Figure 6.

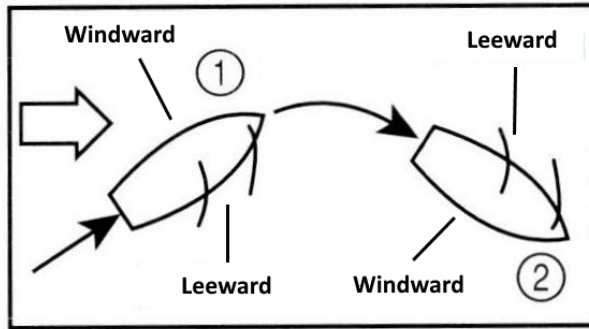


Figure 6: Jib maneuver[9]

2.3 SAILING BEHAVIOR

The sailing behavior is influenced by many factors. The focus of this thesis is on the sailboats changing behavior caused by the aerodynamic and hydrodynamic forces, described in Section 2.1. Especially the case of lee and weather helm, when the sailboat is moving on an upwind course.

2.3.1 Weather helm

Weather helm [5] means that the forces in the sails cause a tendency, which turns the sailboat towards the source of wind. This happens when the lateral pressure point is in front of the sail pressure point. The force $\vec{F}_{LAT}$ is attached to the lateral pressure point and $\vec{F}_T$ is attached to the sail pressure point. As a result a momentum is created which turns the boat towards the wind.

Another reason for a boat to be weather helm is, if the boat has a heeling towards its lee side. The heeling causes a turning of the sail pressure towards the lee side and a turning of the lateral pressure point towards the windward side. The combination between forward thrust and the speed reducing force from the water resistance, turns the boat towards the wind. Both possibilities can be caused by a wrong sail configuration or by a sudden change of the wind direction.

2.3.2 Lee helm

Lee helm [5] means, that the sailboat has a tendency to turn away from the source of wind. This happens when the sail pressure point is in front of the lateral pressure point. As a result there is a momentum which turns the boat away from the wind. The reason for a boat to be lee helm can be caused by a wrong sail configuration or by a sudden change of the wind direction. A schematic demonstration about lee or weather helm can be seen in Figure 7.

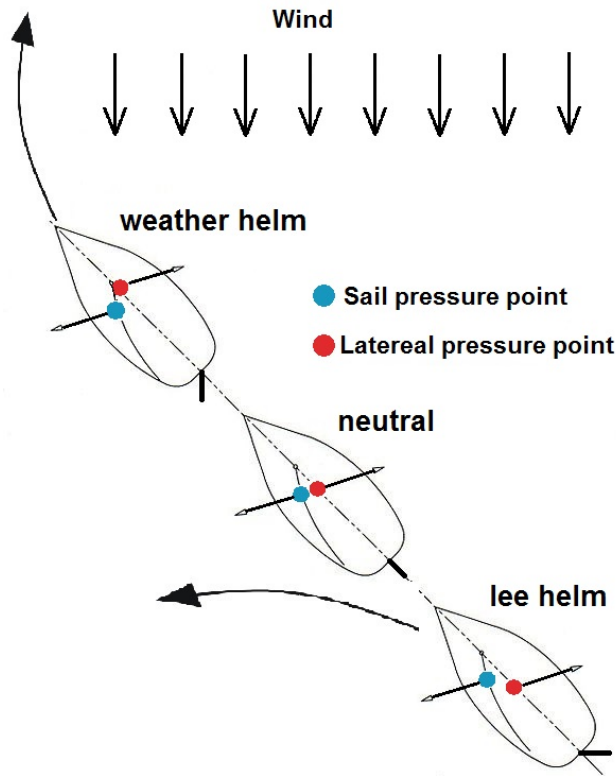


Figure 7: Lee and weather helm behavior[1]

2.3.3 Behavior determination

On a real sailboat, the determination if the boat is either lee or weather helm is a matter of judgment. A sailor feels the difference by monitoring the current pressure on the rudder. When a sailboat is lee or weather helm the rudder must compensate the rotational force by steering against the boat's tendency[47]. This steering causes a hydrodynamic force on the rudder blade and can be felt as a resistance on the steering wheel.

The autopilots control system will do the same by analyzing the rudders angle. In this way the control system determine if the boat is lee or weather helm.

If the rudder has a tendency to steer into the windward side, the boat is lee helm. If the rudder has the tendency to steer into the leeward side, the boat is weather helm. The behavior of lee and weather helm is described in [Figure 7](#) [31].

This chapter introduces the basic physics of a sailboat, which are necessary for this thesis project. It is described how a sailboat uses the wind to produce thrust and the effect of lee and weather helm. Also the possible courses a sailboat can use are also outlined. The next chapter describes how the physical prototype is designed and which electronic devices are used to control the sailboat.

ELECTRONICS

This chapter describes the electronic part of this thesis project and all used electronic devices. The prototype is based on Arduino compatible parts, because the Arduino system allows a fast implementation of the prototype[39].

3.1 SYSTEM DESIGN

The system design contains two different Arduino boards. The first one is the base board for the electronic part for this thesis and contains a Arduino MEGA¹. This Arduino MEGA calculates the wind direction as described in Section 3.3, the current heading angle as explained in Section 3.4 and the current position via GPS as displayed in Section 3.5. Figure 8 illustrates the system architecture and how each component is connected to the system.

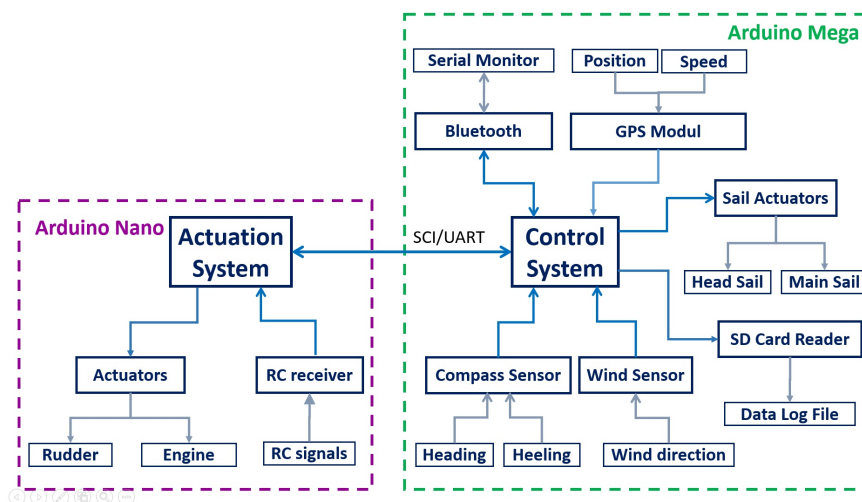


Figure 8: Arduino system architecture

The Arduino MEGA is responsible for the measurement and the calculation which are needed to determine the desired course and the necessary sail angle. To send and observe the whole system during the test phase, a bluetooth module HC05² is installed. With this bluetooth module it is possible to send the course control a new desired course manually and all relevant data can be displayed on a serial monitor.

¹ more detail see: www.arduino.cc/en/Main/ArduinoBoardMega2560

² more detail see: www.electronicaestudio.com/docs/istdo16A.pdf

For evaluation purpose a SD card reader is attached to the arduino mega to log all data which is needed for the system evaluation. In order to reduce the number of cables and to save space inside the boat, some of the modules are soldered to a Arduino MEGA prototype shield.

As shown in Figure 9 the bluetooth module, the GPS module and the SD card reader are used on the prototype shield. This solution does not only save space inside the boat, it also supports to keep order and reduces the risk of electrical failure caused by a loose cable.

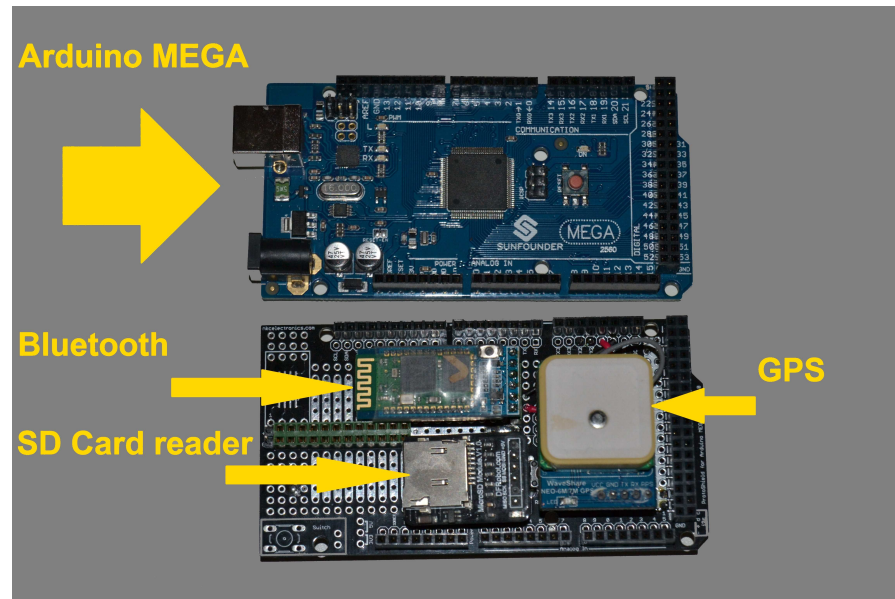


Figure 9: Assembled Protoshield

The second Arduino board is the Arduino NANO³ board. The Arduino NANO board is used as the communication interface between the Arduino MEGA, the back up remote control panel and the servos. The solution with the second Arduino is especially useful during the testing phase, in case of a control system failure or a failure of any measurement system, the boat can be maneuvered back ashore safely. Another part of the system design is the actuating element. As actuators for the sails, two special analog servos are used, for the rudder a digital servo is used. The analog servos have a range of two turns. The digital servo which is used for the rudder, has a range of 120°. The rudder servo is attached to the Arduino NANO fail safe purpose. In the test setup a digital servo is used as rudder actuator, because the rudder movements need to be very precise. Digital servos are more precise and faster compared to analog servos. The exact and precise position of the rudder is important in order to have a stable control system.

³ more detail see: www.arduino.cc/en/Main/ArduinoBoardNano

Another safety feature of the test setup is the installed backup engine, which is also used to maneuver the boat back ashore safely. The backup engine is a high power DC motor with a propeller attached to its shaft. Like the servos, the backup engine is controlled by the Arduino NANO. Figure 10 shows the inside of the physical prototype and all electronic devices.

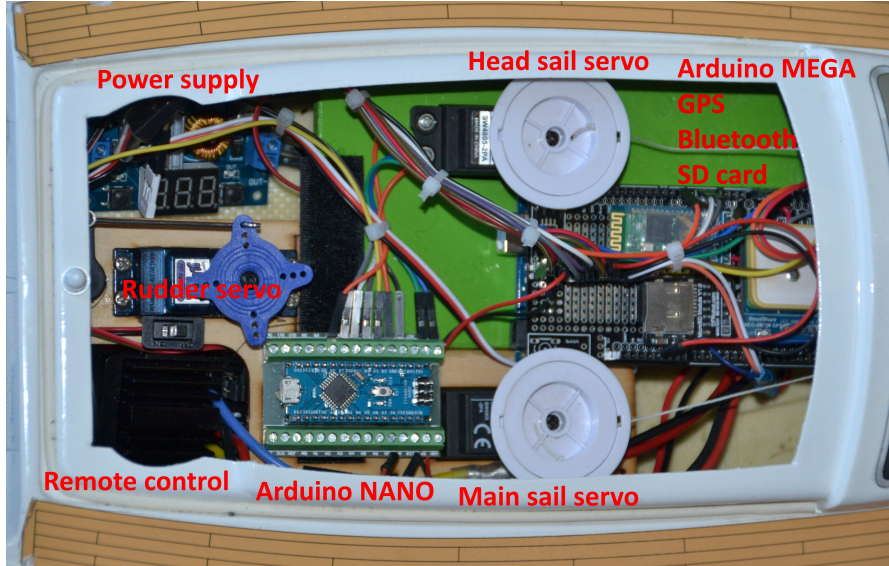


Figure 10: Boat electronics

The system design of the test setup also includes a power supply, it is a lithium polymer battery with 7,5V and 3000mAh and is able to provide up to 120 Amperes. The system setup also includes a DC-DC converter, as the Arduinos and other electronic devices need a 5V power supply. It converts the existing 7.5V constantly to 5V, with a maximum current of 5A. The maximum output power from the DC-DC converter is 25W.

3.1.1 Power consumption

In order to ensure a failsafe power supply on board, it is necessary that the whole system does not run out of power. The calculations must be done for every single energy consumer on board, details about them can be found in Table 2. The table displays all energy consumptions, all electronic components on board and details about their energy consumptions. More details about the maximum power consumption can be found in the data sheet of each device. Every row displays the used voltage in volt, the maximum current in ampere and the resulting power in watt. As described in Table 2, the maximum power consumption given from the data sheets for each component, the overall power is 7.83W.

The total required power is 7.83W and the needed current is 1630mA.

With the existing energy amount fo 3000mAh, the maximum usage can be calculated by using Equation 3.

$$\frac{\text{Energy}}{\text{Current}} = \text{Usagetime} \quad (3)$$

$$\frac{3000\text{mAh}}{1630\text{mA}} = 1.84\text{h} \quad (4)$$

Equation 4 shows the test systems maximum usage time, it is at 1.84 hours or 110.4min.

Table 2: Maximum Power Consumption

Component	Voltage [V]	Current [A]	Power [W]
Arduino Mega	5	0.081	0.405
Arduino Nano	5	0.035	0.175
Wind Sensor	5	0.0005	0.0025
GPS	5	0.1	0.5
SD Card Reader	5	0.1	0.5
Bluetooth	5	0.04	0.02
Compass	5	0.025	0.125
Actuators:			
Rudder	5	0.25	1.25
Main Sail	5	0.5	2.5
Head Sail	5	0.5	2.5
Total:		1.63	7.98

3.2 SERVOS AND ENGINE TRIGGER

The servos used in this thesis project, are commercial available analog and digital servos which are commonly used in remote controlled vehicles. Those servos are triggered by a PWM signal [15]. A PWM signal is a series of repeating pulses of adjustable width.

The period time for servo control is standardized with the value of 20ms. The width of the pulse defines the position of the servo. To move the servo to the neutral position, the pulse width has to be 1.5ms. In order to turn the servo to its minimal position the pulse width has to be 1ms and to turn it to the maximum position the pulse width it has to be 2ms. Figure 11 illustrate the positioning by using a PWM.

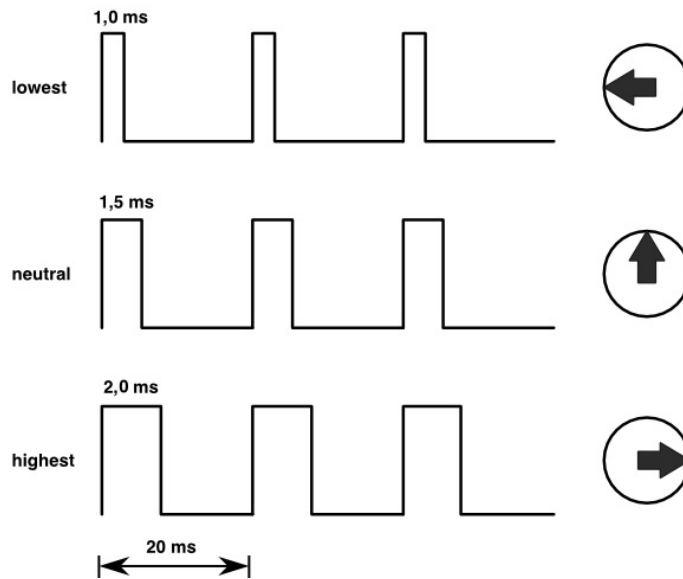


Figure 11: Servo PWM[2]

The digital servo, which is used to move the rudder, has a range of 120° where the minimum is -60° and the maximum is at 60° . For this thesis project the maximum and minimum position for the rudder is limited from -45° to 45° . This is important to avoid a steering overshoot. For the sail actuation two special analog servos with a range of 720° are installed. The rudder and the sail servos are position controlled and directly attached to a digital output from the Arduino NANO.

The used engine is a DC motor which needs a driver circuit because the DC motor needs more power than the electronic devices. The DC motor is attached to the output from the driver circuit. The driver circuit is attached to a digital output of the Arduino NANO. As described in Table 3, the Arduino NANO sends a PWM signal to the driver circuit, which converts it to a DC voltage for the DC motor.

Table 3: Servo and engine trigger

PWM width	Rudder servo	Sail servo	Engine
1.0 ms	left	fully closed (15°)	throttle backwards
1.5 ms	middle	half open (45°)	stop
2.0 ms	right	fully open (90°)	throttle forward

3.3 WIND SENSOR

Commercial available sensors to measure the wind direction are either too big⁴, not precise enough⁵ or very expensive⁶. For this thesis project, a custom-made wind sensor, based on results of the literature research, was built[26, 31]. The wind sensor for this thesis project is based on a 10k Ω precision rotary potentiometer⁷.

The mechanical parts are constructed with SolidWorks⁸ and printed with a PLA filament 3D printer⁹. The wind sensor construction is displayed in Figure 12.

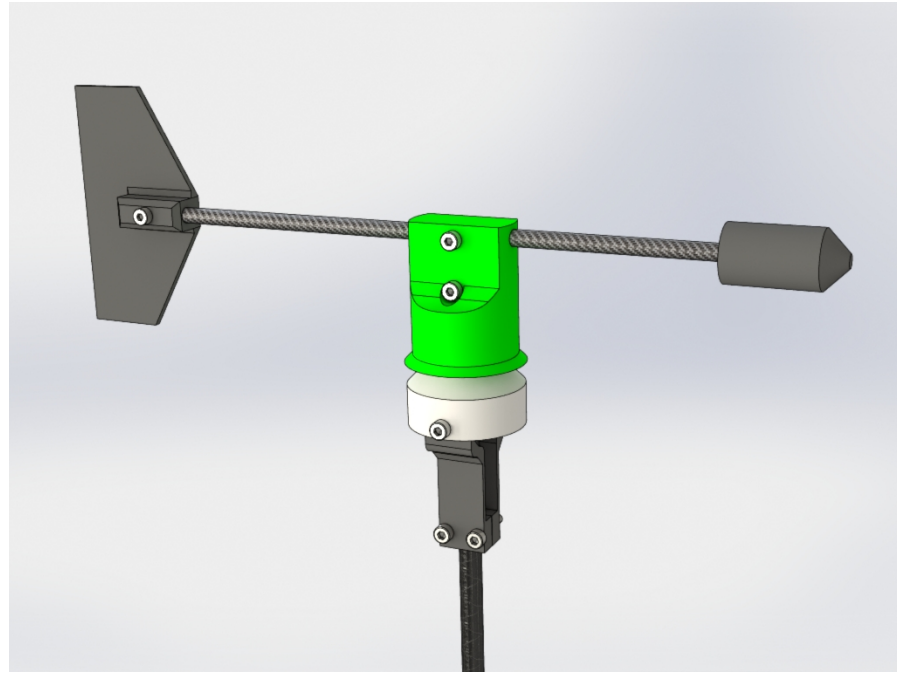


Figure 12: Wind sensor

The measured wind direction is a relative value, if the wind vane is pointing into the heading direction the wind angle is 0°. Because the sail adjustment is only depending on the relative wind direction there is no need to calculate the absolute values for the wind direction. The absolute wind direction value is from 0° to 360°, where 0° is equal the magnetic north. The potentiometer is connected to an analog input on the arduino mega. The circuit is shown in Figure 13.

The analog input measures the apparent applied voltage and provides a value from 0 to 1023. Depending on the potentiometer position the Arduino MEGA measures the current voltage.

4 more detail see: www.aanderaa.com/productsdetail.php?Wind-Direction-Sensor-34

5 more detail see: www.sparkfun.com/products/8942

6 more detail see: www.scientificsales.com/Model-101908-Current-Loop-Wind-Sensors-4-20mA-Out-p/101908.htm

7 more detail see: www.bourns.com/docs/product-datasheets/6539.pdf

8 more detail see: <http://www.solidworks.com/>

9 more detail see: www.ultimaker.com

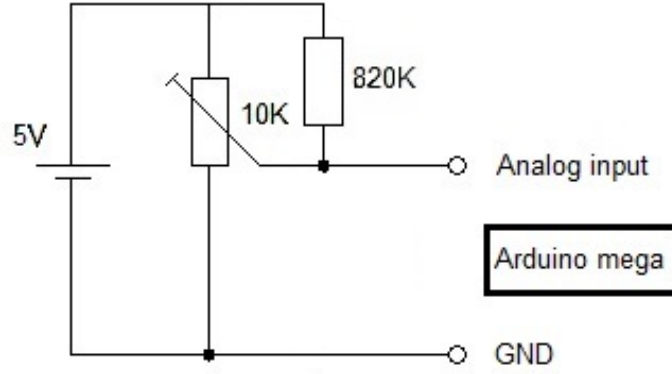


Figure 13: Potentiometer circuit[3]

A full rotation is from 0V to 5V and repeating. In order to prevent a failure within the measurement, which can be caused by a small gap between the 1Ω and 10KΩ, a resistor with 820KΩ is soldered between the positive terminal and the wiper. This resistor ensures that the wiper always has a contact. If the potentiometer is at an undefined position where the wiper has no contact, the Arduino MEGA could not measure any voltage. This scenario can appear when the wiper is between 0° and 360°.

The analog input values are mapped to the maximum potential on the potentiometer. With this information and the Equation 5 the theoretical precision for the wind sensor can be calculated.

$$\frac{\text{Rotation}}{\text{Resolution}} = \text{Precision} = \frac{360^\circ}{1024} = 0.35^\circ \quad (5)$$

3.4 COMPASS AND ACCELERATION SENSOR

For navigating a boat, it is important to know the current heading. On a real sail boat, the current heading is measured with a magnetic compass. Compasses are very old tools for navigation, they were first mentioned in 1040AC. Nowadays, the magnetic compasses are replaced by electronic compasses, which are able to measure the magnetic field from the earth and calculate the current heading. The data given from the electronic compass can be used for autopilots. Those compass modules are basically acceleration sensors with two or three axis.

Another use of the compass and the acceleration sensor, is the measurement of the roll angle. This thesis project uses the roll angle to determine wind gusts or to recognize a too high pressure on the sails.

Commercially available compass modules like the HMC6352¹⁰ or the HMC5883l¹¹ have a accuracy of 1° - 2° , if they have no pitch or roll movement while they are measuring. Whereas if they have a pitch or roll movement during the measurement, they do not compensate the changing angle of the magnetic field. To compensate this angle for the acceleration sensor it is important to use tilt compensation[17, 34] for these acceleration sensors. To reduce wrong heading data, a CMPS11¹² module is installed with the purpose of measuring the current heading. The CMPS11 module has an integrated tilt compensation and is connected to the Arduino MEGA via Inter-Integrated Circuit (I²C). The CMPS11 module is displayed in Figure 14.

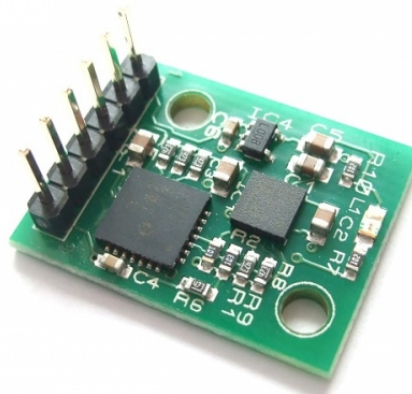


Figure 14: Compass module CMPS11[7]

The sensor location on the boat is very important because the electronic devices cause a lot of Electro Magnetic (EM) interference. The electronic devices, like the engine or the servos, produce a EM field, which manipulates the compass sensor. This is the reason why the CMPS11 sensor is placed at the front section of the prototype, where no metal or electronic device is. To find the location with the least magnetic interference, a small magnetic compass was used. The CMPS11 sensor is placed in a housing, which also constructed with Solid Works. The sensor housing is illustrated in Figure 15.

¹⁰ more detail see: www.sparkfun.com/products/retired/7915

¹¹ more detail see: www.adafruit.com/product/1746

¹² <https://www.robot-electronics.co.uk/htm/cmeps11i2c.htm>

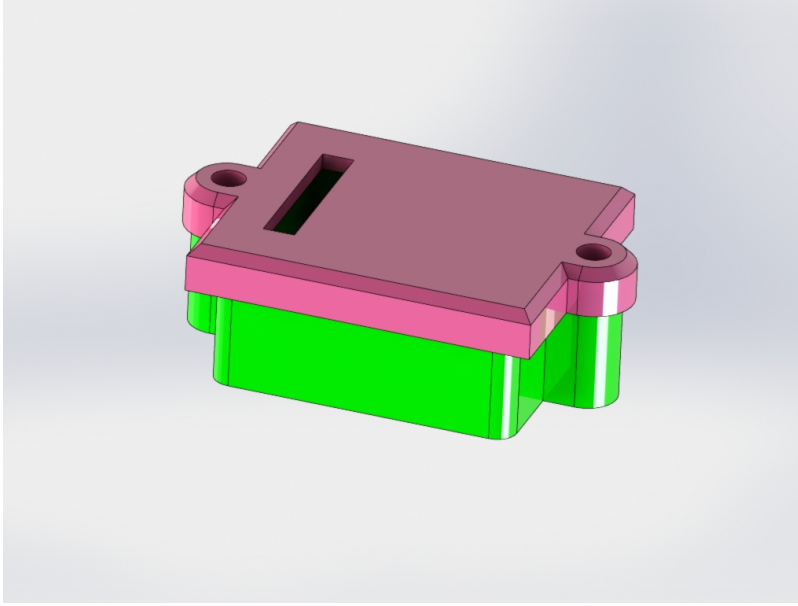


Figure 15: Compass sensor housing

3.4.1 Tilt compensation

To measure the heading with the compass sensor properly, the sensor need to be fixed on its pitch and roll angle. If the roll or pitch angle is not equal to 0° , this causes a failure in the measurement. Even if the sensor is mounted properly on the sailboat, the roll and pitch angles of the sensor are never constantly zero. To reduce the measurement failure of the heading on the sailboat, the compass sensor needs to be tilt compensated[19].

The compass sensor measures the magnetic field from the earth. If the sensor is not precisely leveled to the earth's surface, it can not measure the magnetic field. This is illustrated in Figure 16, the coordinate system with the index C is for the compass module and the coordinate system with the index E is the plane parallel to the earth's surface. The angle ϕ displays the roll angle and the angle θ shows the pitch angle of the compass. The vector $\vec{g}$ represents the earth's gravity direction. In order to measure the right heading angle, the pitch and roll angle need to be compensated [20, 21].

The tilt compensation algorithm is similar to the bearing computation. The bearing calculation is needed for determining the desired course given the current position and the target position. More details can be seen in Section 4.2.1. The tilt compensation algorithm uses the arctangent function to calculate the current heading φ , as displayed in Equation 10. The variables X and Y are the tilt compensated magnetic vectors, illustrated in Equation 8 and Equation 9. These values are compensated by using the roll and pitch angle.

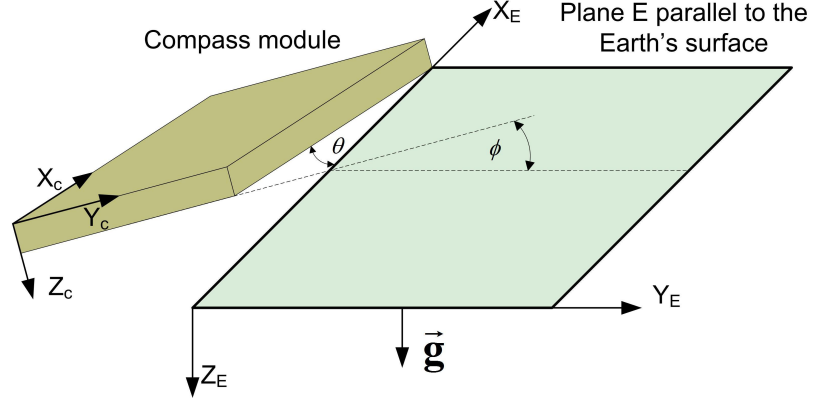


Figure 16: Compass tilt[12]

Equation 6 and Equation 7 are substitutions for A_x and A_y . More detail of the tilt compensation algorithm can be seen in [27].

$$A_x = -\sin \phi \quad (6)$$

$$A_y = \sin \theta \cos \phi \quad (7)$$

$$X = X_c(1 - A_x^2) - Y_c A_x A_y - Z_c A_x \sqrt{1 - A_x^2 - A_y^2} \quad (8)$$

$$Y = y_c \sqrt{1 - A_x^2 - A_y^2} - z_c A_y \quad (9)$$

$$\phi = \arctan \frac{Y}{X} \quad (10)$$

3.5 GLOBAL POSITIONING SYSTEM

In this thesis project the GPS is used to receive the current position and the current speed of the boat. To compute a desired course with the course control system it is necessary to have the current GPS position. With the current GPS position of the sailboat and a target GPS position the desired course can be calculated. This calculation is described in section Section 4.2.2.

In this thesis the boat speed is also measured by using the GPS. The speed is measured with the GPS because it is not possible to install a paddle wheel sensor for speed measurement. Common used paddle wheel sensors for speed measurement on real boats are too big for the sailboat model, used for this thesis project.

For this thesis project a u-blox NEO6¹³ GPS chip is used. The NEO6 has an accuracy of less than 2m. This can be seen in the data sheet of the GPS module.

The NEO6 chip uses the National Marine Electronics Association (NMEA) protocol and is built to receive the Recommended Minimum Sentence C (RMC) messages. RMC is the recommended minimum sentence, which receives only the essential GPS data and reduces the memory usage in of the Arduino MEGA. The received data from the RMC are time, date, position, speed over ground in knots and the course over ground in degree. To retrieve the GPS data from the received NMEA data, the received sentence needs to be parsed and translated. The GPS data is written in a single sentence and separated with semicolon. The GPS module is configured to receive RMC sentences with a ration of 1Hz. Whereas the Arduino MEGA is programmed to trigger an interrupt when there is a new GPS signal available.

3.6 REMOTE CONTROL

The Remote Control (RC) is a backup solution for this thesis project. In case of a failure of the control system or any sensor error, the boat can be maneuvered back ashore, by using the remote control. In recreational RC vehicles the servos are directly attached to the RC receiver. The RC receiver, receives a 2,4Ghz signal from the RC panel and converts it to a PWM signal. The converted values of the RC are used to control the servos.

As described in Section 3.2, the signal value is between 1ms and 2ms. The value of 1.5ms is the middle position for a servo. 1ms is the minimum position and 2ms is the maximum positions of a servo. The outputs from the RC receiver are connected to defined digital inputs on the Arduino NANO board. One of the channels is acting as a switch to activate or deactivate the RC. The Arduino NANO observes this channel. In case of emergency the channel is activated and the Arduino MEGA uses the data from the serial connection which is attached to the Arduino NANO and deactivates the control system to steer the boat back ashore with the RC panel.

¹³ more detail see: www.u-blox.com/de/product/neo-6-series

This chapter provided an overview of the electronics a sailboat needs to gather data. Starting with the system design, an overview for the whole system was described. It was also shown how all components are communicating with each other. Also the power consumption for all electronic devices were introduced, as well as the control of the servos by using a [PWM](#).

Finally all sensors, which are responsible for measuring the wind direction and the heading and the current position, are described.

[Figure 17](#) shows the final physical prototype. It can be seen that the compass sensor is placed in the front of the boat and the wind sensor is mounted on top of the boat.

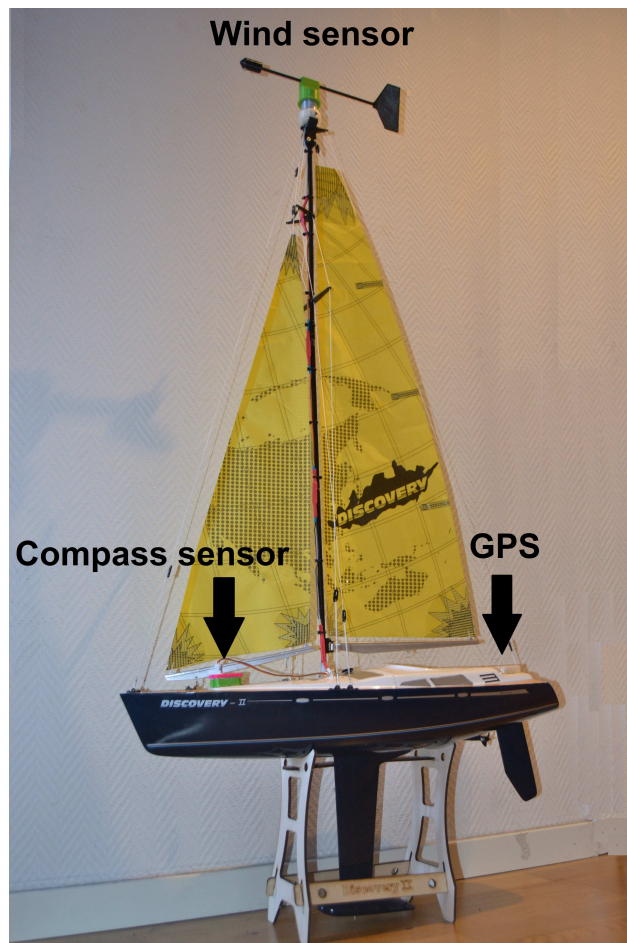


Figure 17: Physical prototype

How the collected data is used for controlling a sailboat automatically, is described in the next chapter.

CONTROL SYSTEM

This chapter introduces the control system of the sailing autopilot. It provides information about the control logic and the details about it. To accomplish this thesis project goal, the control system must be equipped with knowledge from the sailing domain. This chapter shows how the connection is done and how the data is used for the individual steps.

4.1 CONTROL LOGIC

In order to show the feasibility of this thesis, the implementations for the control systems are based on state-of-the-art research on robotic sailboats [28, 37, 44, 49]. Compared to [28], this thesis project uses a Proportional Derivative (PD) controller instead of only a proportional controller for following the desired course. The PD controller has the advantage that the derivative term eliminates the overshooting of the controller.

In contrast to [49], where the control system is wind independent, the control system in this thesis project is able to react on changing wind direction immediately. This is a more realistic approach in order to use it on real sailboat. The control system in this thesis uses machine learning for computing the sail angle, therefore it is independent of mathematical models of a sail boat. In [37], it can be seen how to use mathematical models of the sails and the boat to compute the sail angle.

The control system for this thesis project is split into two different parts. The first part is the course control system, which is responsible for the course calculation and the actuation of the rudder in order to stay on the defined course. The second part is the sail control system, it is responsible for calculating the optimal angle between the sails and the wind direction. The calculations do not only depend on the current course, they also need to consider the direction of the wind stream.

The sailboat has two different types of sail modes: Fully autonomous drive and manual drive. As mentioned in [Section 3.6](#) the Arduino NANO listen to the RC control switch. If this switch is deactivated, the autopilots control system is activated and the boat is sailing fully autonomously.

4.2 COURSE CONTROL

The most essential part for a sailboat autopilot is the course control. It is important to be able to follow a desired course. In this thesis project a [PD](#) controller is implemented. More detail to the controller is described below in [Section 4.2.2](#).

The current course on a real sailboat, can be influenced by the hydrodynamic drift caused from a water stream and from the aerodynamic drift caused by the sail forces. In order to reach a defined target position, it is important to compensate the drift and calculate the desired course in real time.

4.2.1 Course finding

In this thesis project, the calculation of the desired course $\bar{\theta}$ is done by the autopilot automatically. The desired course depends on the current [GPS](#) position $[\varphi_1, \lambda_1]$ and the target position $[\varphi_2, \lambda_2]$, where φ is the lateral point and λ is the lateral point. The input for the course calculation is the target position.

Caused by the fact that the earth is not flat, a straight course on the sea chart can not be mapped to a straight line in the actual navigation. For example, a boat travels from Cape Town ($33^\circ 55' 21.6''$ S, $18^\circ 25' 0.08''$ E) to Melbourne ($37^\circ 48' 0''$ S, $144^\circ 57' 0''$ E) the desired course at the beginning would be $\sim 141^\circ$ and changes along its way to $\sim 42^\circ$. This means that the desired heading needs to be recalculated at every new [GPS](#) position [[16](#), [41](#)]. The principal of changing bearing is illustrated in [Figure 18](#).

To calculate a course from Cartesian coordinates, [Equation 11](#) uses the Arctangent2 function, which is the arctangent function with two arguments. Since the Arctangent2 function returns a value between -180° and $+180^\circ$ the output angle needs to be converted to a bearing angle.

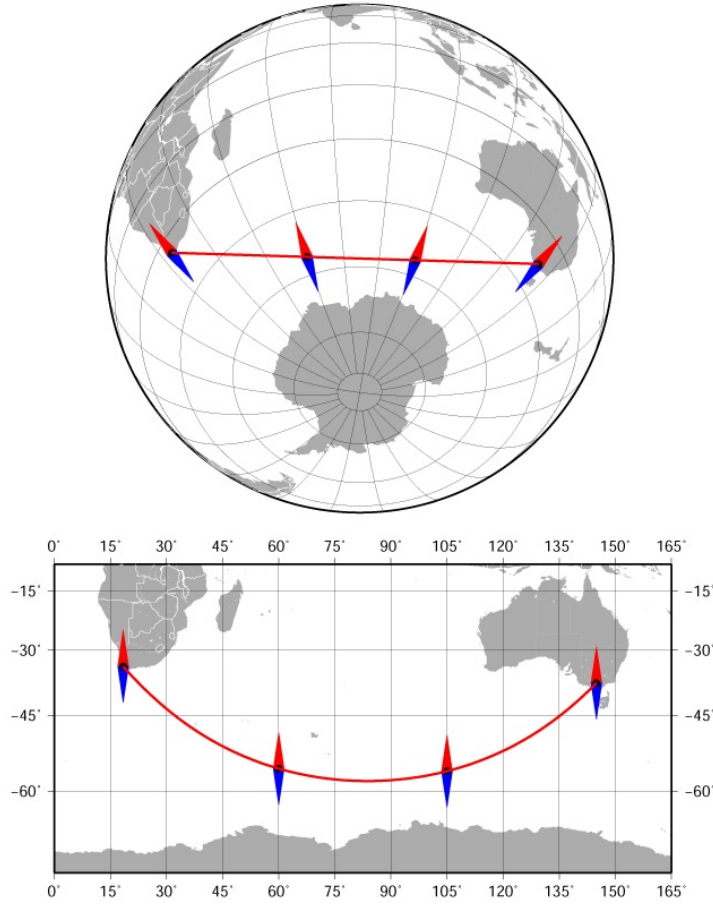


Figure 18: Changing bearing example[6].

If the output angle from the Arctangent2 function is negative, 360° is added to the value. Equation 13 and Equation 14 calculate the values for the Arctangent2 function, by using the current position and the target position. Equation 11 calculates the longitudinal difference between the current and the target position. Equation 12 shows the calculation for the longitudinal difference [8].

Variable declaration:

φ_1 – Lateral position 1,
 φ_2 – Lateral position 2,
 λ_1 – Longitudinal position 1,
 λ_2 – Longitudinal position 2,
 $\Delta\lambda$ – Longitudinal difference.

$$\bar{\theta} = \arctan 2(y, x) \quad (11)$$

$$\Delta\lambda = \lambda_2 - \lambda_1 \quad (12)$$

$$x = \cos(\varphi_1) * \sin(\varphi_2) - \sin(\varphi_1) * \cos(\varphi_2) * \cos(\Delta\lambda) \quad (13)$$

$$y = \sin(\Delta\lambda) * \cos(\varphi_2) \quad (14)$$

4.2.2 Heading controller

One of the most important tasks for a sailboat autopilot is to keep the boat on the desired course. This is triggered by the rudder, the autopilot steers it automatically. To follow a course from 0° to 360° , the control system must control the rudder position σ_R . The input value Θ is provided by the compass module, as described in [Section 3.4](#). The current heading error θ_e is the difference between the current heading θ and the desired heading $\bar{\theta}$.

In this thesis project, a [PD](#) controller is used for the closed control loop, to control the rudder position. In [\[22, 45\]](#) the heading controller is implemented as a Proportional Integral Derivative ([PID](#)) controller. The [PD](#) controller has no overshoot compared to the [PID](#) controller because there is no integrative part. In this thesis project the desired course is also calculated when there is a new [GPS](#) signal available. This is also a new approach compared to [\[22, 45\]](#). The control loop for this thesis project is shown in [Figure 19](#)[\[23\]](#).

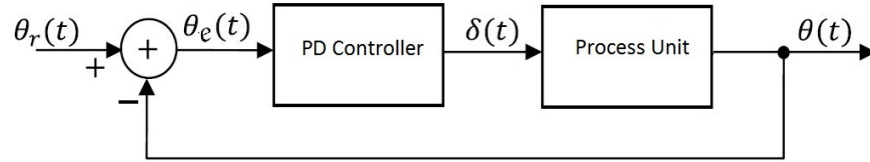


Figure 19: Closed loop for heading control

A [PD](#) controller consists of two different stages. The Proportional section produces an output value, which is proportional to the error value θ_e . Whereas the differential section, calculates the derivative error θ_e , by multiplying the rate of change by the derivative gain [\[32\]](#). For this thesis the equations for the proportional is calculated as shown in [Equation 15](#).

$$\theta_e(t) = \theta(t) - \bar{\theta}(t) \quad (15)$$

$$\sigma_R(t) = K_P * \theta_e(t) + K_D \frac{d\theta_e(t)}{dt} \quad (16)$$

To be able to use the equation of the PD controller for the system identification, it is necessary to transform the equation from the time domain into the frequency domain. This can be done with the Laplace transformation[38]. After the Laplace transformation from the Equation 16, the transfer function for the PD controller is shown in Equation 17.

$$R(s) = K_p * (1 + T_v s) \quad (17)$$

The values for K_p and T_v are determined empirical. To tune the PD controller, the Ziegler Nichols method is used[25]. The Ziegler Nichols method is an empirical method to tune the parameters for the PD controller, the parameters are calculated in Equation 18 and Equation 19.

$$T_v = 0.15 * T_{critical} = 0.15 * 1.108 = 0.166334 \quad (18)$$

$$K_p = 0.5 * K_{p,critical} = 0.5 * 10.5078 = 5.2539 \quad (19)$$

Within this section, it was shown how to implement the closed control loop with a PD controller. The next section describes the identification for the process unit, which is displayed in Figure 19.

4.2.3 System identification

In order to be able to do simulations for the system evaluation, it is necessary to identify the system's process unit in this system. The identification for the whole system is done with the System Identification Toolbox provided by MATLAB¹. The model identification is based on the data out of the experiments with the boat. To identify the system, MATLAB needs the input and output data from the system to calculate the transfer function. The input and output data is taken by the logged data on the SD card. With the input and output data the Identification Toolbox is able to simulate the overall system. Figure 20 shows the overall output data in red and the simulated model in blue.

¹ more detail see: www.mathworks.com/products/sysid.html

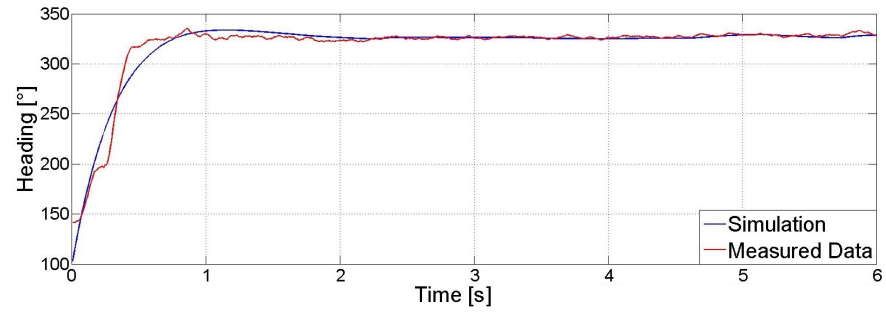


Figure 20: Course control simulation

Figure 21 shows the simulated frequency response for the Amplitude and Figure 21 shows the simulated frequency response for the phase.

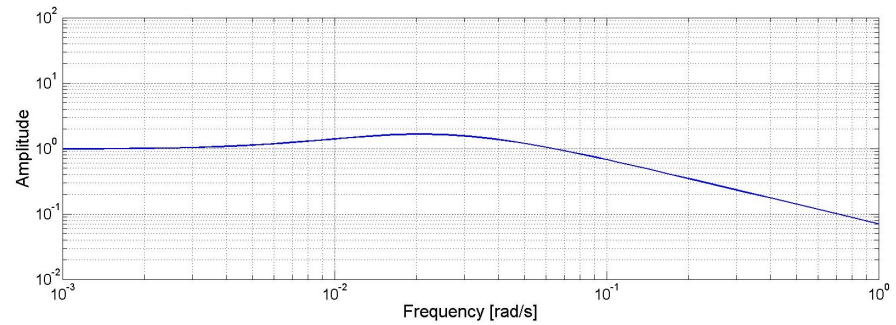


Figure 21: Frequency response (Amplitude).

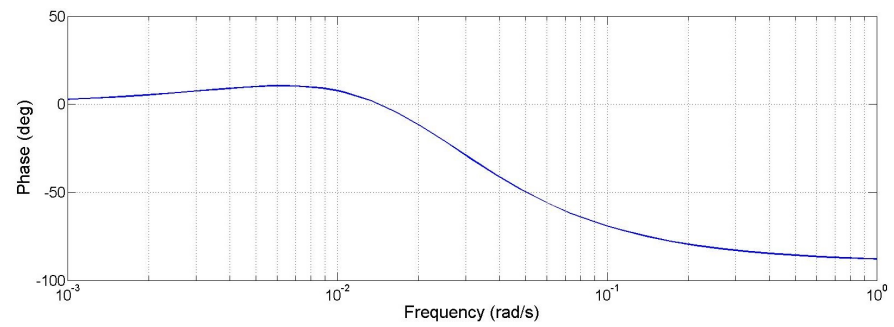


Figure 22: Frequency response (Phase).

The overall transfer function $m(s)$ for this thesis project, is shown in Equation 20.

$$M(s) = \frac{s + 0.00763}{s^2 + 0.0446s + 0.0005474} \quad (20)$$

In order to retrieve the transfer function for the process unit $G(s)$, the equation for the closed control loop $M(s)$, shown in Equation 21, can be transformed to Equation 22. The solved and final equation with all values is shown in Equation 23.

$$M(s) = \frac{R(s)G(s)}{1 + R(s)G(s)} \quad (21)$$

$$G(s) = \frac{M(s)}{R(s) - M(s)R(s)} \quad (22)$$

$$G(s) = \frac{1}{K_p} * \frac{s + 0.00763}{s^3 - 0.1533s^2 - 0.15328s + 0.00069115} \quad (23)$$

4.3 SAIL CONTROL

The sail control system controls the angle of the sails. The foundation for the sail control system, is a lookup table, containing static values for the sail angle depending on the current wind direction. The wind direction is measured by the wind sensor as described in Section 3.3. Figure 23 describes the basic sail angle according to the current wind direction.

When the behavior of the boat is changing, the sail control system determines the type of change and reaction to different situations. When the heeling of the boat increases dramatically, because of a sudden wind gust, the both sail angles are opening to reduce the pressure on the sails during the heeling. After the heeling is stabilized again the sail angle is closed back to the original angle.

If the boat is lee helm, the sail angle of the main sail is closing by 1° . This is repeated every second until the boat behavior is in the neutral state again.

If the sailboat is weather helm, the sail angle for the main sail is opening by 1° per second, until the boat is in the neutral state again. The sails angle changes by adjusting their offset value σ_{SHO} for the head sail and σ_{SMO} for the main sail.

The neutral state is reached, when the boat is neither lee or weather helm, nor is the heeling angle too big. Table 4 describes how many degree the offset values for the sails are tuned, depending on the different situations.

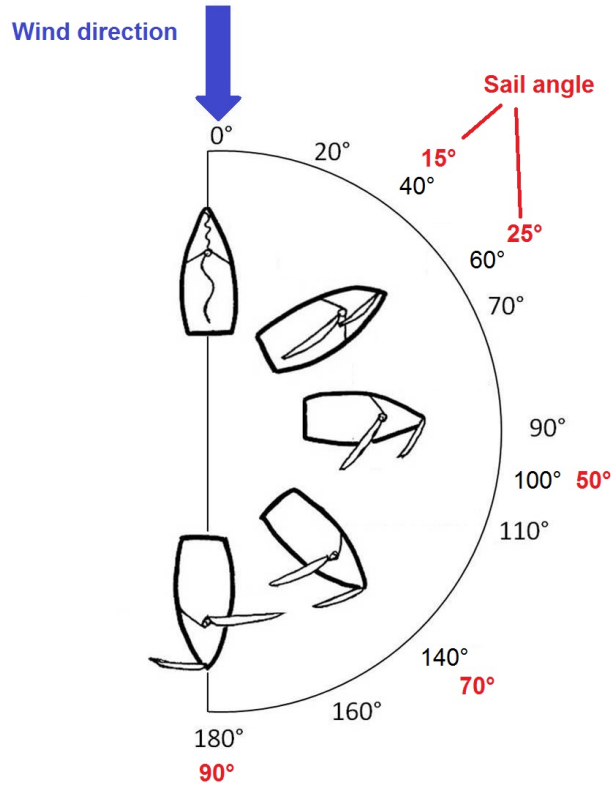


Figure 23: Schematic sail angles

Table 4: Behavior determination and offset values

	σ_R	Roll Angle	σ_{SHO}	σ_{SMO}
Lee Helm	$>5^\circ$	-	-	-1°
Weather Helm	$<-5^\circ$	-	-	$+1^\circ$
Heeling	-	$45^\circ > \sigma_R < -45^\circ$	$+10^\circ$	$+10^\circ$
Neutral	$5^\circ < \sigma_R > -5^\circ$	$45^\circ < \sigma_R > -45^\circ$	-	-

The sail control system is implemented as a mealy state machine, because it changes its state depending on the current state and the input data[18, 46]. The functionality for each state is described below the whole system is displayed in Figure 24.

S1:

S1 is the neutral state where the control system learns the optimal angle for the sails. The boat is in neutral position when its behavior is within the limits, which are defined in Table 4. If the boat turns to lee helm behavior, the state S2 gets activated, whereas if the boat turns to weather helm behavior, state S3 is activated. But if the boat turns to a high heeling angle, state S4 is activated.

S₂:

S₂ is the lee helm state, where the sail control system increases the pressure on the main sail. This is done by reducing the offset value σ_{SMO} by 1° per second, until the boat is in the neutral state again and S₁ is activated.

S₃:

S₃ is the weather helm state, where the sail control system decreases the pressure on the main sail. This is done by increasing the offset value σ_{SMO} by 1° per second until the boat is in the neutral state again and S₁ is activated.

S₄:

S₄ is the heeling state, where the sail control system decreases the pressure on both sails. This is done by increasing the offset value σ_{SMO} and σ_{SHO} by 10° per second until the boat is in the neutral state again. Before leaving this state, the offset values are reset to the value before the boat was heeling. Once the boat is back in the neutral position, state S₁ is activated.

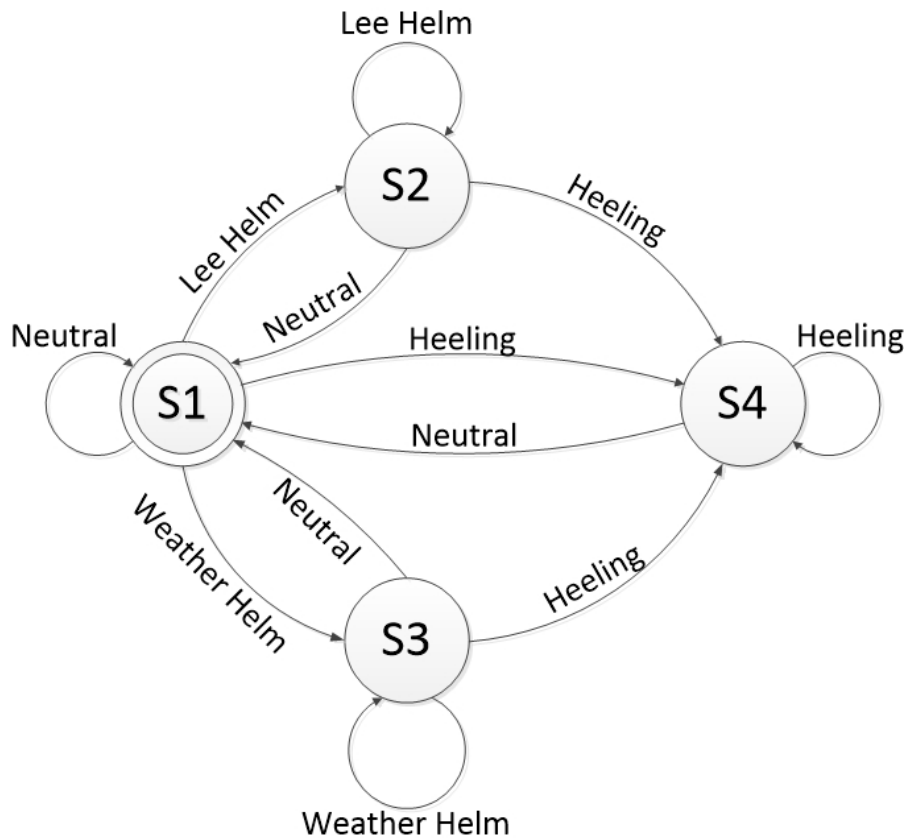


Figure 24: State machine of the sail control system

Table 5 shows how the state machine determines which state is going to be next, depending on the current state and the input data. The first column shows the current state, the last column shows the next possible step, depending on different combinations of the input data. The input data are information about the current sailboat behavior of the sailboat and the current heeling from the acceleration sensor. More detail can be found in Section 4.3.2.

4.3.1 Wind direction

For the behavior determination, it is important to define the windward side and the leeward side of the boat. This information is needed for the behavior determination. If the relative wind direction is between 0° and 180° , the boats starboard side of the boat is windward and the boats port side of the boat is leeward. Whereas if the relative wind direction is between 180° and 360° the boats port side of the boat is windward and the boats starboard side of the boat is leeward.

Table 5: State machine table

Current State	Lee Helm	Weather Helm	Neutral	Heeling	Next State
S1	-	-	-	✓	S4
S1	-	-	✓	-	S1
S1	-	✓	-	-	S3
S1	✓	-	-	-	S2
S1	-	✓	-	✓	S4
S1	✓	-	-	✓	S4
S2	-	-	-	✓	S4
S2	-	-	✓	-	S1
S3	-	-	-	✓	S4
S3	-	-	✓	-	S1
S4	-	-	✓	-	S1

4.3.2 Behavior calculations

The average rudder position σ_R measured for a defined time frame of two second. The results are used to determine if the sailboat is lee helm or weather helm. The signs describes if the rudder is steering towards the wind or away from the wind. The values for the behavior determination are displayed in table Table 4.

Lee helm:

When σ_R had steered the rudder away from the wind source with an average of 5° over the last two seconds, the boat behavior is set to lee helm.

Weather helm:

Is σ_R steering the boat towards the wind source with an average on -5° over the last two seconds, the boat behavior is set to weather helm.

Heeling:

When the heeling or roll angle of the boat increases to a value higher than 45° or lower than -45° , it means the pressure on the sail is too high. This can be caused by a wind gust or a not optimal sail angle.

4.3.3 *Learning algorithm for sail angle*

While the state machine, described in [Section 4.3](#), has activated the neutral state S_1 , the sail control system tries to find the best sail angle in order to optimize the boat speed. Therefore the system uses a decision tree, which is displayed in [Figure 25](#). Compared with [\[44\]](#), where the sail angle depends on the heeling angle of the boat, this thesis project uses the current speed of the boat in order to compute the sail angle. Compared to [\[24, 37\]](#) the use of a decision tree also reduces the processing power for the Arduino MEGA.

This system searches the maximum speed, comparable to a hill climbing algorithm and decides with an decision tree how to increase the boat speed [\[36\]](#). The decision tree decides in each cycle, if the boat speed is increasing or decreasing and sets a new offset value for the sail angle.

As displayed in [Figure 25](#), there are two different decisions. Increasing means that the sail pressure on both sails is increasing by closing the sail angle by 1° . Decrease means that the pressure on both sails is decreasing by open the sail angle by 1° .

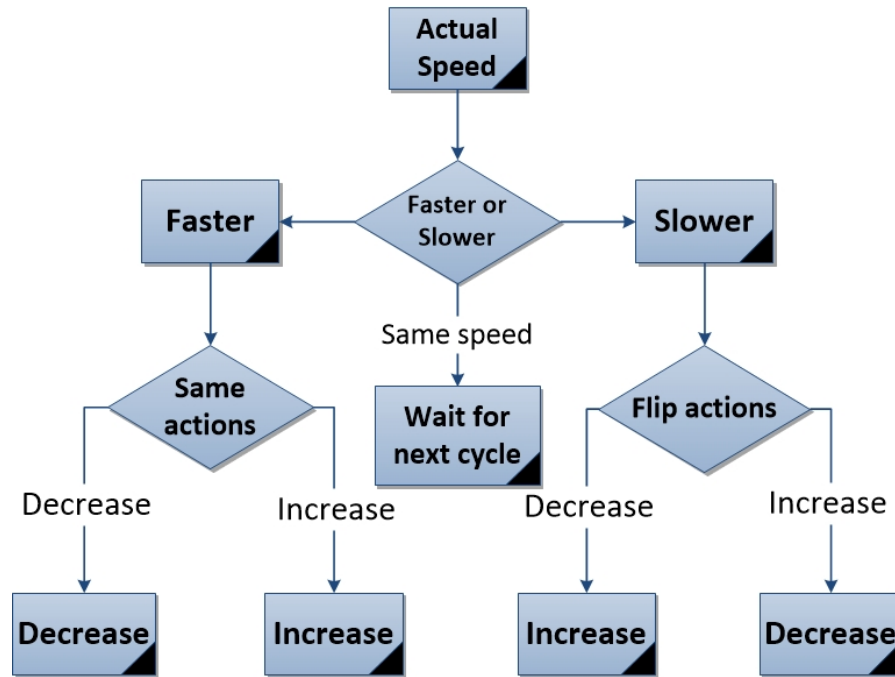


Figure 25: Sail angle decision tree

Each cycle of the decision tree is event triggered by the [GPS](#) interrupt, which is described in [Section 3.5](#). With each new available [GPS](#) speed value, the decision tree is executed. The first decision is made by comparing the speed value of the old cycle with the speed value of the new cycle.

The new decision is made by observing if the new speed is higher, lower or equal to the speed from the previous cycle. In the case of equal speed values, the system waits for the next cycle. Every change for the offset value is defined by 1° . The second decision depends on the previous action.

- In leaf of increased speed, the new decision is the same decision than in the previous cycle. In the case that the previous decision was to increase the pressure, the new decision is also to increase the pressure. The action for increasing the pressure is illustrated in [Figure 26](#).
- In the leaf of decreased speed, the new decision is the opposite decision than in the previous cycle. This means if the previous decision was to increase the pressure on the sails, the new decision is to decrease the pressure. The action of decreasing the pressure is illustrated in [Figure 27](#).

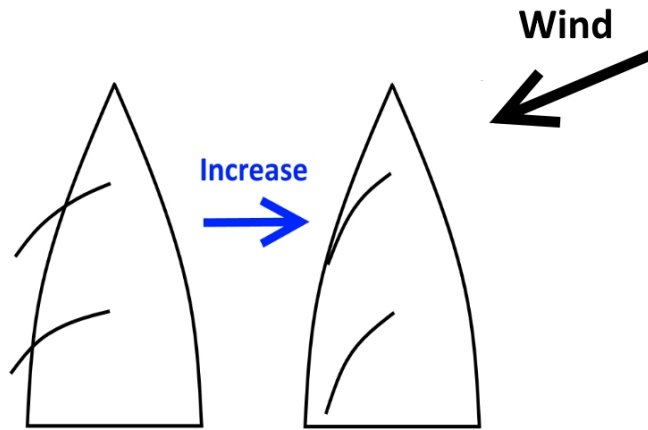


Figure 26: Increasing the pressure on the sails

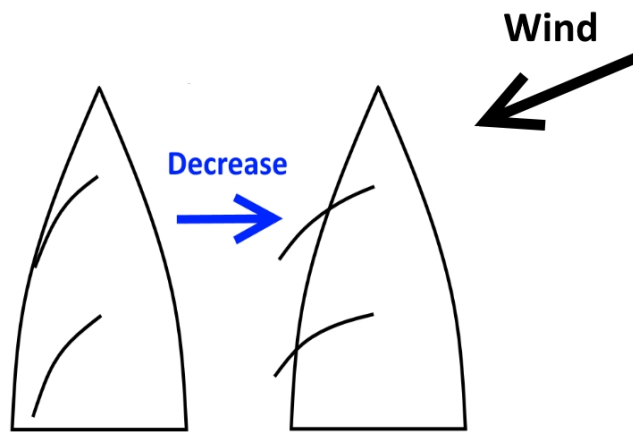


Figure 27: Decreasing the pressure on the sails

The previous chapter gave a detailed insight into the control system, which was implemented for this thesis project. Therefore the course control system was split up into the course finding, the heading control and the system identification. The results from the system identification can be used for further simulations for the physical prototype, which was constructed during this thesis project.

The sail control system was explained with a finite state machine. The input values for the state machine are the different behaviors of lee helm, weather helm and the heeling angle of the boat. These input values are also described in this chapter.

The next chapter describes all results, which were determined during the experiments.

EXPERIMENTAL RESULTS

The experiments took place at the *torvsjön* lake near Halmstad. This lake has good conditions for the experiments because it is not too big, which is good to keep the boat within remote control distance. But it is also not too small, which is necessary to be able to log all needed data for the evaluation of this thesis project.

Therefor the boat sailed from the start point with the coordinates $56^{\circ}41'54.1''$ N $13^{\circ}00'18.4''$ E to the end point with the coordinates $56^{\circ}42'03.1''$ N $13^{\circ}00'05.8''$ E. The sailing distance was 211m and the desired course at the starting point is 324.374° . The sailing course was tracked with GPS and illustrated in Figure 28. This illustration was generated with Google Earth¹

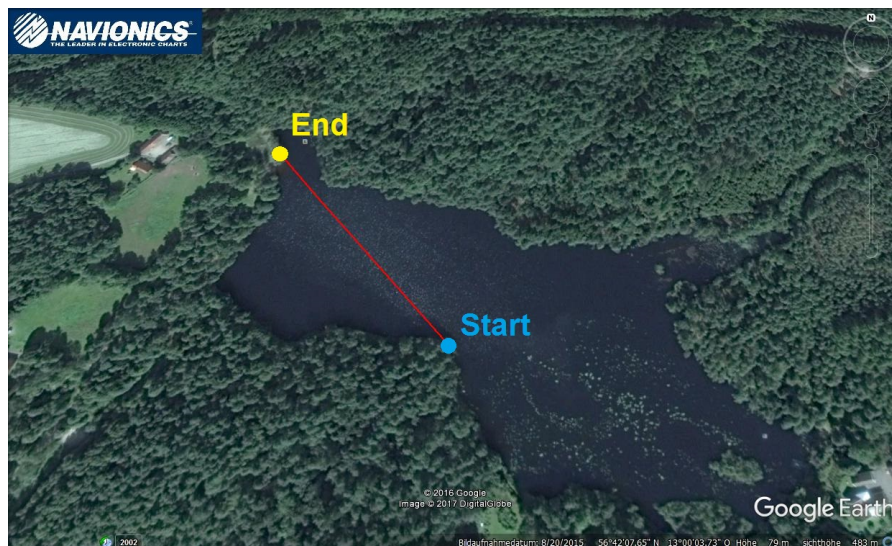


Figure 28: Desired course from the experiment

The accuracy for the control system was determined with the provided data from the GPS, the compass sensor, the wind sensor and all calculated data from the control system. The data is logged on the SD card. A simulation model is determined of the collected data from the experiments, as described in Section 4.2.3. All collected data is analyzed and plotted with MATLAB².

The accuracy of the course control system is measured with GPS data and a simulated comparison from the determined model, with another course control system model from [22].

¹ more detail see: www.google.com/intl/en/earth/

² more detail see: www.mathworks.com

The sail control system is analyzed with the internal data given from the Adruino MEGA and from the changing boat speed and heeling angle.

5.1 COURSE CONTROL RESULTS

The accuracy of the course control system is measured with the GPS data, which was stored on the SD card. Figure 29 shows the recorded GPS track, compared to the desired course. The red line illustrates the desired course from the starting point to the end point whereas the blue line stands for the measured GPS track.

The measured maximum deviation from the desired course is 3.28m. As the GPS module has an accuracy of less than 2m, in the best case, all data from the GPS track are in the range of the maximum GPS accuracy. Due to the reason that the GPS has an accuracy of 2m and the maximum deviation is 3.28m, the accuracy of the system must be higher than the measurement error from the GPS. This is already a good value for the small scaled prototype. If this system would be installed on a real sail boat, this accuracy would be sufficient.

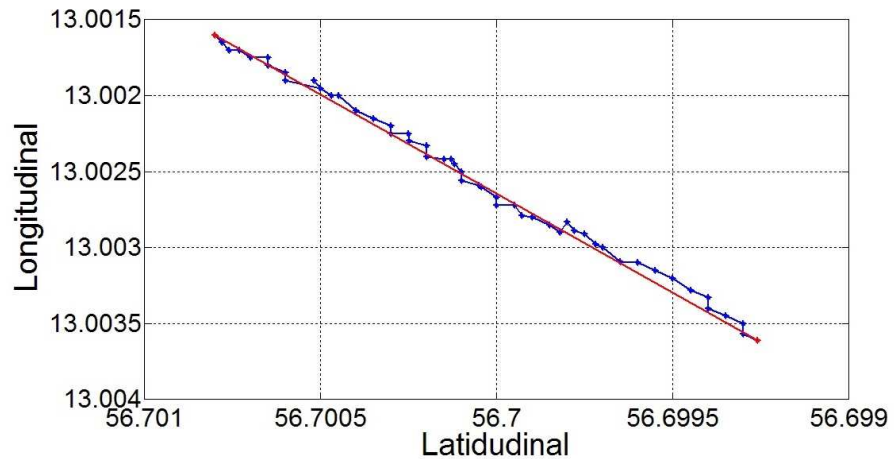


Figure 29: Desired course and GPS track.

5.1.1 Course control simulations

For the simulations of the course control system the mathematical model for the process unit was calculated, as described in Section 4.2.3. With this model it is possible to compare the accuracy, of the course control system, with simulations of other systems. Figure 30 shows the measured data in red and the simulated data in blue. It can be seen that the system has a low overshoot and similarities to a PT₁ element. It is also noticeable that there are two remarkable peaks during the course changing.

These peaks are small deviations of the current heading caused by some small waves during the course changing of the boat. It can be seen that if the boat has reached its desired course, that the waves have less impact to the current heading.

All simulations within this thesis are implemented with MATLAB Simulink³.

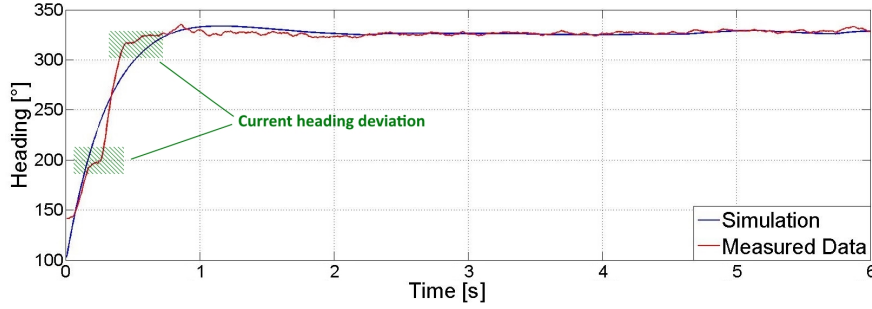


Figure 30: Measured and simulated output.

To have a comparable step response the control system of a participant from the world robotic sailing championship was consulted. Based on the paper [22] the transfer function for the process unit $G(s)$ is shown in Equation 24.

As a controller for this system, a Proportional Integral (PI) controller was implemented. The transfer function for the controller $R(s)$ is shown in Equation 25 [14, 33]:

$$G(s) = \frac{0.34}{s(s + 0.7)(s + 0.6)} \quad (24)$$

$$R(s) = 0.38 * \frac{s + 0.005}{s} \quad (25)$$

It can be seen, that the difference between those two systems is the overshoot in the step response. This is mainly caused by the different controllers. The controller in this thesis project is a PD controller which has less overshoot than the other system, which uses a PI controller.

³ more detail see: www.mathworks.com/products/simulink.html

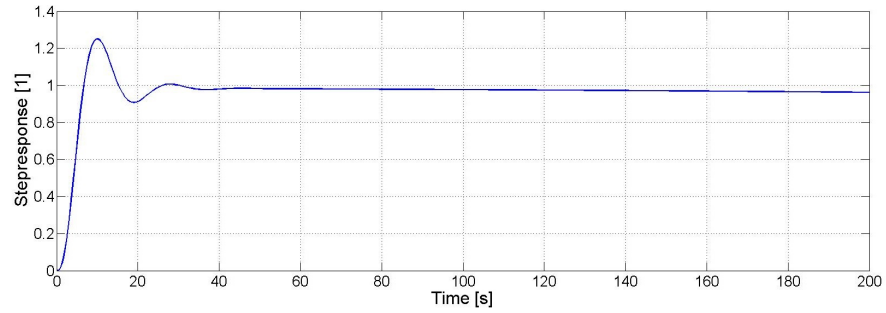


Figure 31: Other system simulation.

5.2 SAIL CONTROL RESULTS

To measure the accuracy of the sail control system, it is important to look at some different aspects. The aspects are, the boat speed depends on the current sail angle, the sail angle optimization in order to increase the boats speed and the observation of the heeling angle to get information about the pressure in the sails.

The boat speed depends on the current sail angle which gives some quality measure about the sail angle optimization as described in [Section 4.3.3](#). The sail angle depending on the current wind direction. This gives information about the reaction on a changing wind direction and the current sail angle depending on the current heeling angle. The heeling angle is used to determine the accuracy of the reaction to a changing pressure on the sails caused by wind gusts.

5.2.1 Sail optimization

In order to measure the accuracy of the learning algorithm which computes the sail angle, the logged data from the boat speed is compared with the current sail angles from head and main sail. The boat speed comes from the [GPS](#) module and the sail angles are calculated values from the sail control system.

The [Figure 32](#) shows some huge leap values, which are physically impossible for a sailboat. The measured leap in the speed data appears, because of the combination of few speed and the inaccuracy of the [GPS](#) module. If the boat sails with a speed of 1.5kn or equal to 0.75 meters per second, and with the information that the 2m accuracy of the [GPS](#), the measured speed can not be accurate. This means that the boat speed is not provided optimally by the [GPS](#) is not optimal for a small scaled sail boat.

A possible solution for this problem can be the use of a water stream sensor. The same system is used on real sailboats. On a real sailboat this water stream sensor is mounted underneath the water line.

The sensor can measure the boat speed by measuring the stream through the water.

Although the test setup was not able to measure the speed with the planned accuracy, the functionality can be shown by comparison. Comparison of the speed data illustrated in Figure 32 and the changing sails angle data for the had and main sail shown in Figure 33.

The sail angle is a value from 0° , which means that the sails are fully open, to 90° , which means that the sails are fully closed.

The two plots show, that every time the speed is increases or decreases, the sail angle for both sails are also changing. This simultaneous changed sail angle describes that the system works as expected.

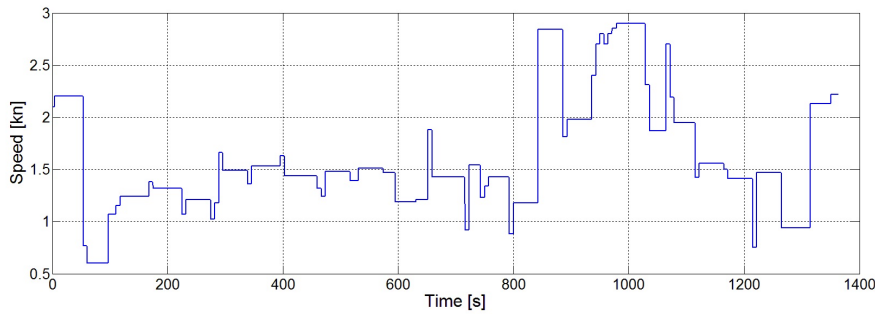


Figure 32: Boat speed.

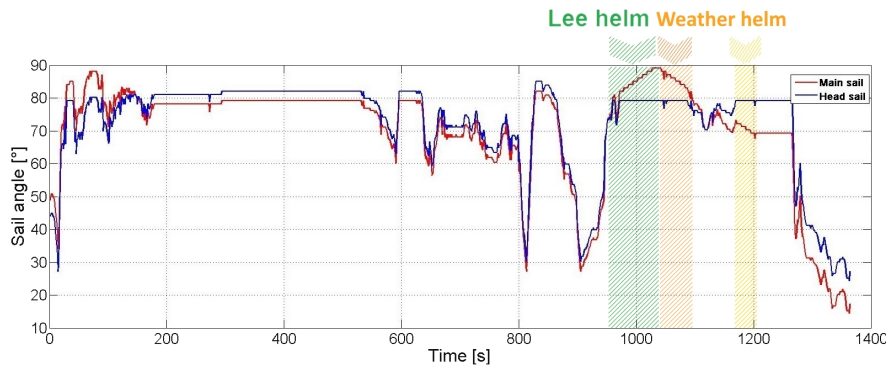


Figure 33: Sail angle.

5.2.2 Lee and weather helm control

Figure 33 shows not only the changing sail angle depending on the speed, it also illustrates the sail angle comparison between the main sail and the head sail. With that comparison it can be clearly seen that the sail control system reacts on the defined behavior of lee and weather helm.

In the green marked section, the main sail is closing its angle until the boat turns into weather helm, which is displayed in the orange section. Afterwards, the boat needs to do another weather helm maneuver in order to get into the neutral state again, this can be seen in the yellow marked section.

5.2.3 Sail angle and wind direction

During the experiment, the boat follows not only the desired course, it also reacts on changing wind conditions. Figure 34 illustrates the current wind direction during the experiment. When those two figures, Figure 34 and Figure 33 are compared with each other, it can be seen clearly that the system reacts on the changing wind conditions automatically.

At the end of the experiment, at time 1200s, it can be seen that the sailboat changes its course from an upwind course to a downwind course. During this course change, also the sails are also adjusted to the changed wind direction.

This sail angle adjustment is important to be able to stay on the desired course.

Exactly this behavior, the automated sail adjustment, is the main difference to already existing sailboat autopilots. And it is also the main advantage for sailors itself.

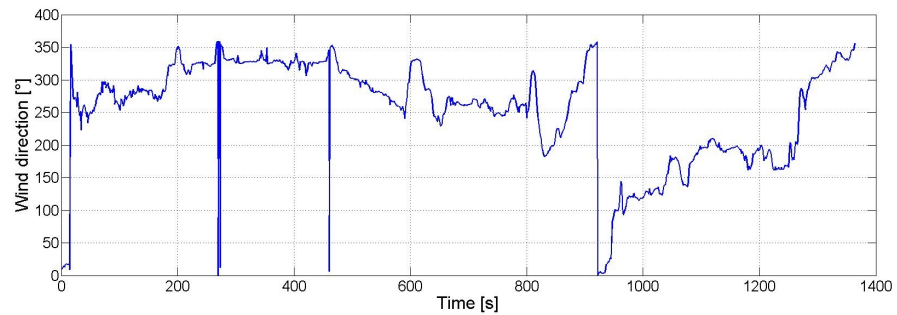


Figure 34: Wind direction.

5.2.4 Sail angle and heeling

In order to be able to react on wind gusts or on the situation of too much pressure on the sails, it is necessary to observe the heeling angle of the boat. On a real boat this could be done with a wind speed sensor. In this thesis project the compass sensor, as described in [Section 3.4](#), is used to determine wind gusts, by observing the heeling angle of the boat. [Figure 35](#) displays the angle for the main sail and the head sail and the heeling angle. Every time the heeling angle is pushed down to an angle higher than 45° or lower than -45° , the sail control system opens the sail angle. The sail angle is opened by 10° per second until the heeling angle is in normal range again. This is a useful feature in order to avoid a capsize of the boat.

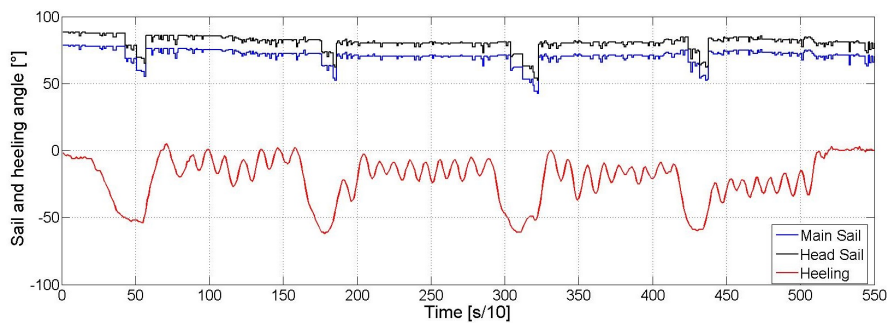


Figure 35: Sail and heeling angle.

This chapter illustrated the results and the feasibility of this thesis. There is still some room for improvements, especially with the speed measurement and the [GPS](#) position estimation. Some ideas, like an increased accuracy of the system or a more usable interface for real sailboats are described in the next chapter.

CONCLUSION AND FUTURE WORK

The final chapter of this work will summarize the work done and will give an overview of the results of this thesis project as well as an outlook to possible future work.

The target of this thesis was to show the feasibility of a sailboat autopilot with automated sail actuation. The accuracy of the system has been measured by analyzing the logged GPS track compared to the desired course. The result for the course control system evaluation is, that a PD controller is feasible for following the desired course. The accuracy of the system is also satisfiable, due to the reason that the maximum deviation to the desired course is 3.28m and the accuracy of the GPS is about ~ 2m.

The sail control system is able to react on changing sailing conditions. It has the ability to distinguish between lee or weather helm behavior, while the boat is sailing on an upwind course.

With the wind sensor data the sail control system can adjust the sail angle if the wind direction changes. Is the heeling angle of the boat too big, the sail control system reduces the pressure on the sails automatically by opening the sail angle. This can because of a wind gust or too much pressure on the sails. The sail optimization is a process where the sail control system observes the speed of the boat.

During the evaluation phase of this thesis project, it turned out that the speed measurement for the sail optimization, with the GPS module is not ideal for a small scaled sailboat prototype. It can be seen that the system is working as expected, but the speed data from the GPS is too volatile to get a satisfiable result. The problem seems to be the slow pace of the boat and the vague accuracy of the GPS. This means there is still room for improvements, which results into a possible task for future work.

A possible future step to improve the accuracy of the GPS and also the quality of the measurement, is to use a Kalman filter [30]. A Kalman filter can be used as a tool for position and speed estimation based on GPS [40]. The use of a Kalman filter can increase the accuracy of the GPS up to less than 0.5m.

One possible enhancement for the sail optimization is to improve the accuracy of the speed measurement. There are two different approaches.

The first one is the use of a paddle wheel sensor which measures the water flow underneath the boat to calculate the boat speed. This is also a common approach to measure the boat speed on a real sailboat. The other one is to improve the [GPS](#) accuracy as described above. An general approach for improving the course control, is to extend the system with a wave compensation[24]. If a sailboat has to sail against waves and the system does not compensate them, the waves can change the course of the boat.

To compensate the waves, some kind of wave estimation could be calculated. The compass sensor already delivers the needed data for roll and pitch angle of the boat. Wave compensation could increase the accuracy of the course control system. It would reduce the small peaks during the course change as described in [Section 5.1.1](#) another possible future step is the maneuver control. This would push the sail control to the next level, would be the maneuver control. It would be a nice feature for the sailboat autopilot to determine when it is necessary to execute a tack or jib maneuver.

Once the autopilot is implemented on a real sailboat, there is another important feature which needs to be added, the collision avoidance. Therefore the course control system would need to import the sea chart for calculating the course. Using them, the control system can avoid areas with shoals or shallows. To avoid collisions with other boats the course control system could be equipped with a radar system. With the radar system, the control system could determine if the boat is heading to a collision with another boat. In the future this thesis can also be used as a platform for studying different problems in control.

Over all this thesis project worked as expected, it is demonstrated that automated sail control is feasible for sailboat autopilots. The course control system and the sail control system are working together as expected. There is still room for improvements, especially with the speed measurement. This thesis project is a foundation to push the idea of automated sail control to the next level, to an autopilot on a real boat.

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